Zhastay Yeltay

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Summary

M.Sc. Computer Science student and Research Assistant at the University of Bonn, working on **generative models for human motion** (text-to-motion) and the **robustness of LLM multi-agent systems**. Equal-contribution first author at **Findings of ACL 2026** and co-author of an award-winning paper at **ACM Web3D 2026**, plus 9 earlier papers on NLP for Kazakh, a low-resource language (39 citations, h-index 5). **5+ years** of industry and research engineering: real-time analytics pipelines on Databricks (data latency 6h → 30m), Spark/SQL optimization on 500M+ rows (**25% faster**), and Kazakh NLP models (accuracy **91% → 97%**).

Education

University of Bonn — **Bonn, Germany**
M.Sc. Computer Science — *Oct 2024 – Mar 2027 (expected)*
- Research on adversarial robustness of LLM multi-agent systems with Lucie Flek's group (b-it, Lamarr Institute), published at Findings of ACL 2026.

Mälardalen University — **Sweden**
M.Sc. studies, Computer Software Engineering — *Jan 2023 – Jun 2023*

Al-Farabi Kazakh National University — **Almaty, Kazakhstan**
B.Sc. Information Security Systems - 1.1/1.0 German Grade (3.84/4.0) — *Sep 2017 – Jul 2021*

Experience

University of Bonn — **Bonn, Germany**
Research Assistant (WHF), Visual Computing Group — *Feb 2026 – Present*
- Design and refine prompts for **text-to-motion** models to improve the emotional quality and semantic alignment of generated motion.
- Work on emotion-based motion classification, generation, and editing.
- Co-authored **BONSAI** (ACM Web3D 2026, **Scientific Excellence Award**), which converts high-capacity neural Gaussian avatars into compact, browser-native streaming representations.
- Optimized a paper project website for mobile, improving performance **3–4x** with no loss in quality.

EPAM Systems — **Remote (Newtown, Pennsylvania, US)**
Data Software Engineer — *Nov 2023 – Oct 2024*
- Built a near–real-time analytics pipeline on **Databricks** with **Azure Data Lake & Delta Lake**, reducing data availability latency from 6h to 30m by automating ingestion and aggregation.
- Optimized **Spark + SQL** jobs on 500M+ rows, cutting runtimes by **25%** through partitioning, Z-ORDER, and broadcast joins.
- Increased nightly workflow reliability by **30%** by implementing idempotent API ingestion with retries and exponential backoff.
- Enabled executives to identify top 5 VIP customer regions and other metrics by designing advanced reporting queries, improving marketing ROI by **12%**.

Al-Farabi Kazakh National University — **Almaty, Kazakhstan**
Researcher & ML/NLP Engineer — *Jul 2019 – Jul 2024*
- Improved Kazakh NLP model accuracy from **91% → 97%** by fine-tuning embeddings and augmenting 200K+ samples.
- Reduced custom stemmer runtime from **4:47 → 0:17 per 1K words**, boosting throughput by **16x** for real-time text pipelines.
- Co-authored **10 publications** (IEEE, Springer, ACM, PeerJ), mostly on NLP for Kazakh, building datasets for destructive-content detection, offensive text classification, and spelling correction.
- Built an AI exam proctoring system with **PoseNet**, detecting cheating with **92% accuracy** in 500+ test sessions.
- Developed backends for ML systems using **Python/Django**, **C#/.NET**, **Java/Spring Boot**, and integrated APIs for data-driven applications.

Otbasy Bank — **Almaty, Kazakhstan**
Software Engineer — *Jul 2022 – Feb 2023*
- Automated 10+ manual processes via **Python RPA**, saving **120 staff-hours/month** and cutting costs by **20%**.
- Decreased back-office workload by **85%** by developing Django-based workflow platform integrated with REST + RPA services.
- Cut customer request turnaround from 2 days to 4h by deploying Selenium + Win32 automation scripts.

Alatau City Bank (ex. Jusan Bank) — **Almaty, Kazakhstan**
Data Engineer — *Oct 2021 – Jul 2022*
- Accelerated ETL jobs by **20%** and reduced infra costs by **30%** through workflow redesign in **Informatica PowerCenter**.
- Delivered high-volume bonus distribution workflows (100K+ monthly records), cutting partner payout errors by **10%**.
- Improved executive reporting speed by **40%** by automating AMARA analytical reports, enabling faster market expansion decisions.

Skills

Programming: Python, SQL, Java (Spring Boot basics), C#
ML/AI: PyTorch, TensorFlow, scikit-learn, vLLM, Ollama, NLP, LLM agents, text-to-motion
Big Data: Apache Spark, Databricks, Kafka, Delta Lake, Hadoop, Hive
Cloud/DevOps: Azure, GCP, AWS, Docker, Git, CI/CD
Languages: English (B2), German (A2), Kazakh, Russian

Selected Publications

1. Alavi, K.*, **Yeltay, Z.***, Flek, L., Karimi, A. (2026). *More Agents Improve Math Problem Solving but Adversarial Robustness Gap Persists.* In *Findings of the Association for Computational Linguistics: ACL 2026*, pp. 43457–43475. (*equal contribution) ACL Anthology | arXiv:2511.07112 | Code

2. Kitz, S., Katz, S., Spindler, J., **Yeltay, Z.**, Rastegari, R., Kaltheuner, J., Stotko, P., Klein, R. (2026). *BONSAI: Browser-Optimized Neural Gaussian Avatar Streaming.* In *Proceedings of the 31st ACM International Conference on 3D Web Technology (Web3D '26)*, Doha, Qatar, pp. 1–11. ACM. **Scientific Excellence Award.** doi:10.1145/3822516.3834888

3. Bolatbek, M., Sagynay, M., Mussiraliyeva, S., **Yeltay, Z.** (2025). *Detection of offensive content in the Kazakh language using machine learning and deep learning approaches. PeerJ Computer Science*, 11:e3027. doi:10.7717/peerj-cs.3027

Other Peer-Reviewed Papers

4. Mussiraliyeva, S., Bolatbek, M., Baisylbayeva, K., **Yeltay, Z.** (2025). *Multi-ideology Dataset for Detecting Online Extremism in Kazakh Language.* ICITA 2024, LNNS vol. 1248, pp. 499–509. Springer. doi

5. Bolatbek, M., Sagynay, M., Mussiraliyeva, S., **Yeltay, Z.** (2024). *Machine learning methods for identifying destructive texts in the Kazakh language.* ADMIT 2024, pp. 99–103. ACM. doi

6. Mussiraliyeva, S., Baisylbayeva, K., Bolatbek, M., **Yeltay, Z.** (2024). *Decoding Ideology: Machine Learning-Based Detection of Extremist Content.* ICCNS 2024, pp. 114–120. IEEE. doi

7. Bolatbek, M., Sagynay, M., Mussiraliyeva, S., Baisylbayeva, K., **Yeltay, Z.** (2024). *Kazakh Language Dataset for Hate Speech Detection on Social Media Text.* IEEE ICCIA 2024, pp. 94–98. doi

8. Mussiraliyeva, S., Bolatbek, M., **Yeltay, Z.**, Azanbay, K. (2024). *Development of an error correction algorithm for Kazakh language.* KazNU Bulletin: J. of Mathematics, Mechanics and Computer Science, 123(3). doi

9. Mussiraliyeva, S., Bolatbek, M., Sagynay, M., Zhumakhanova, A., **Yeltay, Z.**, Medetbek, Z. (2023). *Identifying Cyber-Threatening Texts in the Kazakh Segment of Web Resources.* ICAAI '23, pp. 68–72. ACM. doi

10. Mussiraliyeva, S., Bolatbek, M., **Yeltay, Z.**, Bagitova, K. (2023). *The concept of extremist information and a systematic review of counter-extremism projects* (in Kazakh). Academic Scientific Journal of Computer Science, pp. 112–130.

11. Mussiraliyeva, S., Omarov, B., Bolatbek, M., Ospanov, R., Baispay, G., Medetbek, Z., **Yeltay, Z.** (2021). *Applying Deep Learning for Extremism Detection.* ICAICR 2020, CCIS vol. 1393, pp. 597–605. Springer. doi

12. Tasmurzayev, N.M., Amangeldy, B.S., Nurakhov, E.S., Mukhanbet, A.A., **Yeltay, Z.** (2021). *Implementation of an Intelligent Control System for Heat Distribution in Rooms.* IEEE SIST 2021. doi